



SECTION 14

ORDERING INFORMATION AND MECHANICAL DATA

This section contains the pin assignments and package dimensions of the MC68030. In addition, detailed information is provided to be used as a guide when ordering.

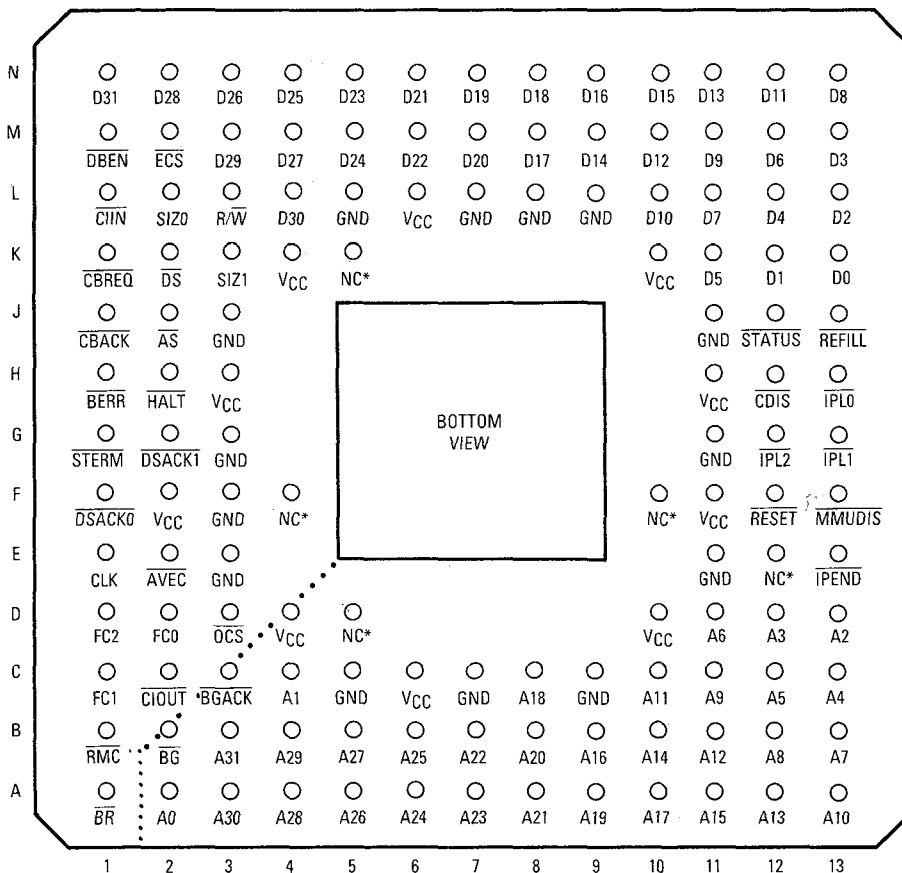
14.1 STANDARD MC68030 ORDERING INFORMATION

Package Type	Frequency (MHz)	Temperature	Order Number
Pin Grid Array RC Suffix	20.0	0°C to 70°C	MC68030RC20
	25.0	0°C to 70°C	MC68030RC25
	33.33	0°C to 70°C	MC68030RC33
Ceramic Surface Mount FE Suffix	20.0	0°C to 70°C	MC68030FE20
	25.0	0°C to 70°C	MC68030FE25
	33.33	0°C to 70°C	MC68030FE33

14.2 PIN ASSIGNMENTS — PIN GRID ARRAY (RC SUFFIX)

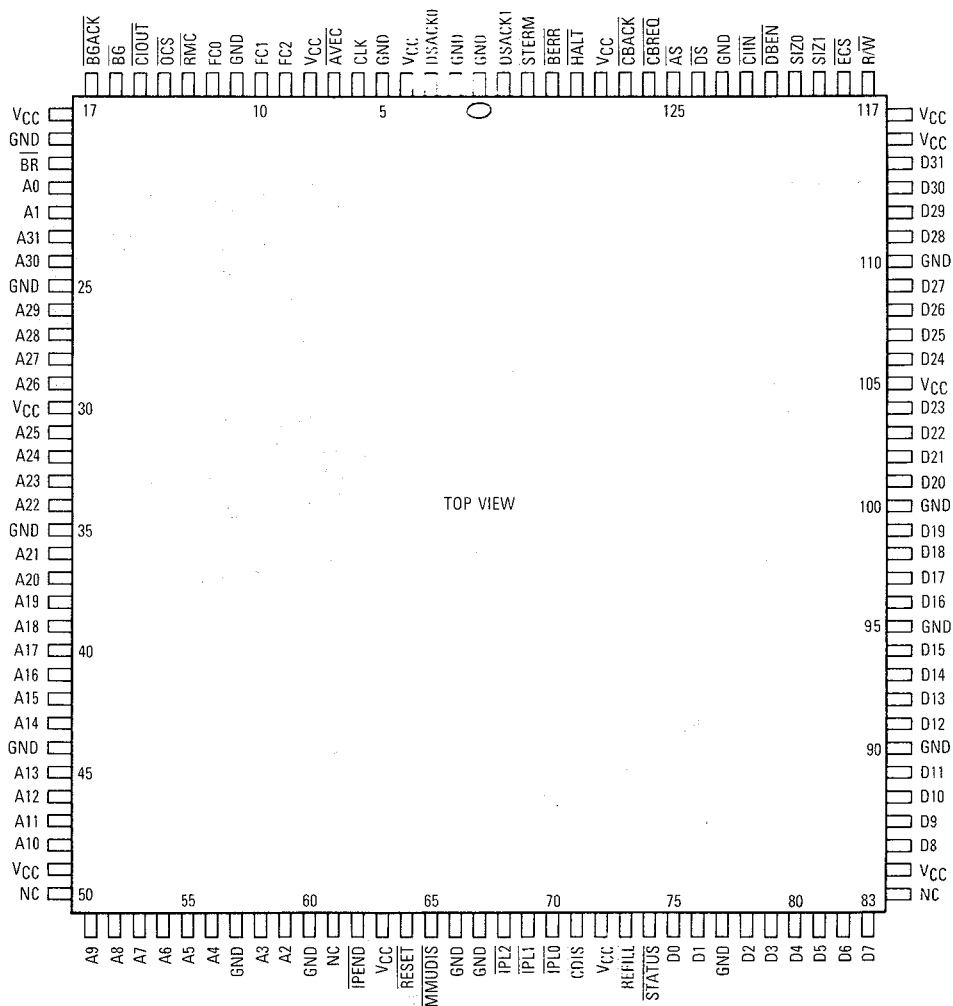
The V_{CC} and GND pins are separated into three groups to provide individual power supply connections for the address bus buffers, data bus buffers, and all other output buffers and internal logic.

Pin Group	V _{CC}	GND
Address Bus	C6, D10	C5, C7, C9, E11
Data Bus	L6, K10	J11, L9, L7, L5
$\overline{\text{ECS}}$, SIZx , $\overline{\text{DS}}$, $\overline{\text{AS}}$, $\overline{\text{DBEN}}$, $\overline{\text{CBREQ}}$, $\text{R}/\overline{\text{W}}$	K4	J3
FC0-FC2 , RMC , $\overline{\text{OCS}}$, $\overline{\text{CIOUT}}$, BG	D4	E3
Internal Logic, $\overline{\text{RESET}}$, STATUS , REFILL , Misc.	H3, F2, F11, H11	L8, G3, F3, G11



*NC — Do not connect to this pin.

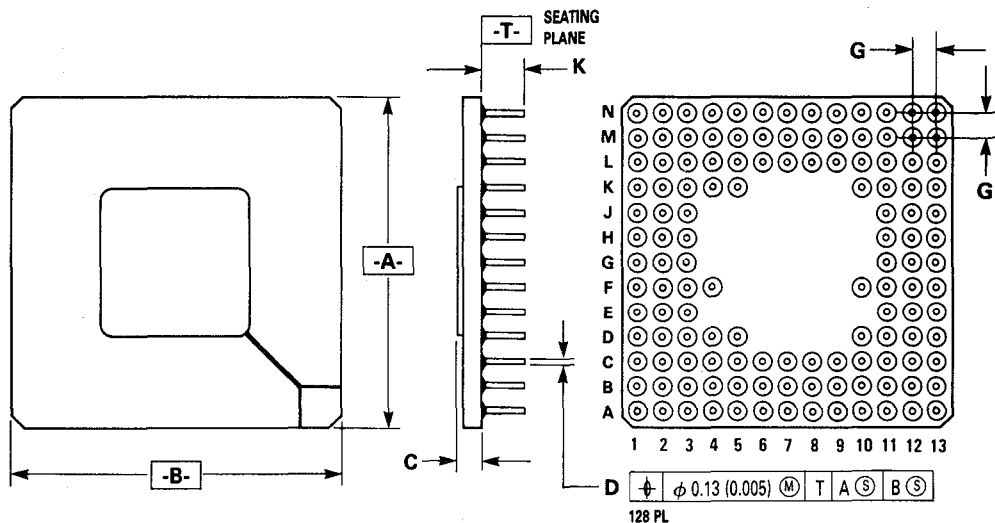
14.3 PIN ASSIGNMENTS — CERAMIC SURFACE MOUNT (FE SUFFIX)



*NC — Do not connect to this pin.

14.4 PACKAGE DIMENSIONS

MC68030
RC Suffix Package
Case 789C-01

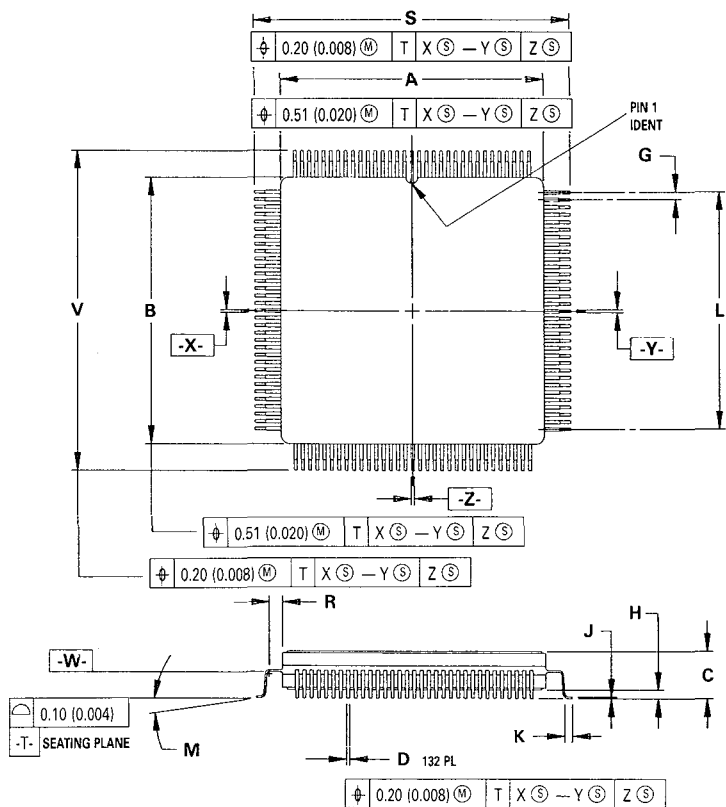


NOTES:

1. A AND B ARE DATUMS AND T IS A DATUM SURFACE.
2. DIMENSIONING AND TOLERANCING PER Y14.5M, 1982.
3. CONTROLLING DIMENSION: INCH.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	34.04	35.05	1.340	1.380
B	34.04	35.05	1.340	1.380
C	2.54	3.81	0.100	0.150
D	0.44	0.55	0.017	0.022
G	2.54 BSC		0.100 BSC	
K	4.32	4.95	0.170	0.195

MC68030
FE Suffix Package
Case 831-01



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	21.85	22.86	0.860	0.900
B	21.85	22.86	0.860	0.900
C	3.94	4.31	0.155	0.170
D	0.204	0.292	0.0080	0.0115
G	0.64 BSC		0.025 BSC	
H	0.64	0.88	0.025	0.035
J	0.13	0.20	0.005	0.008
K	0.51	0.76	0.020	0.030
L	20.32 REF		0.800 REF	
M	0°	8°	0°	8°
R	0.64	—	0.025	—
S	27.31	27.55	1.075	1.085
V	27.31	27.55	1.075	1.085

NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. DIM A AND B DEFINE MAXIMUM CERAMIC BODY DIMENSIONS INCLUDING GLASS PROTRUSION AND MISMATCH OF CERAMIC BODY TOP AND BOTTOM.
4. DATUM PLANE -W- IS LOCATED AT THE UNDERSIDE OF LEADS WHERE LEADS EXIT PACKAGE BODY.
5. DATUMS X-Y AND Z TO BE DETERMINED WHERE CENTER LEADS EXIT PACKAGE BODY AT DATUM -W-.
6. DIM S AND V TO BE DETERMINED AT SEATING PLANE, DATUM -W-.
7. DIM A AND B TO BE DETERMINED AT DATUM PLANE -W-.

